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Kim et al.

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(54) **COOLING DEVICE AND PROCESS FOR COOLING DOUBLE-SIDED SIP DEVICES DURING SPUTTERING**

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C23C 14/50 (2006.01)
C23C 14/34 (2006.01)
(Continued)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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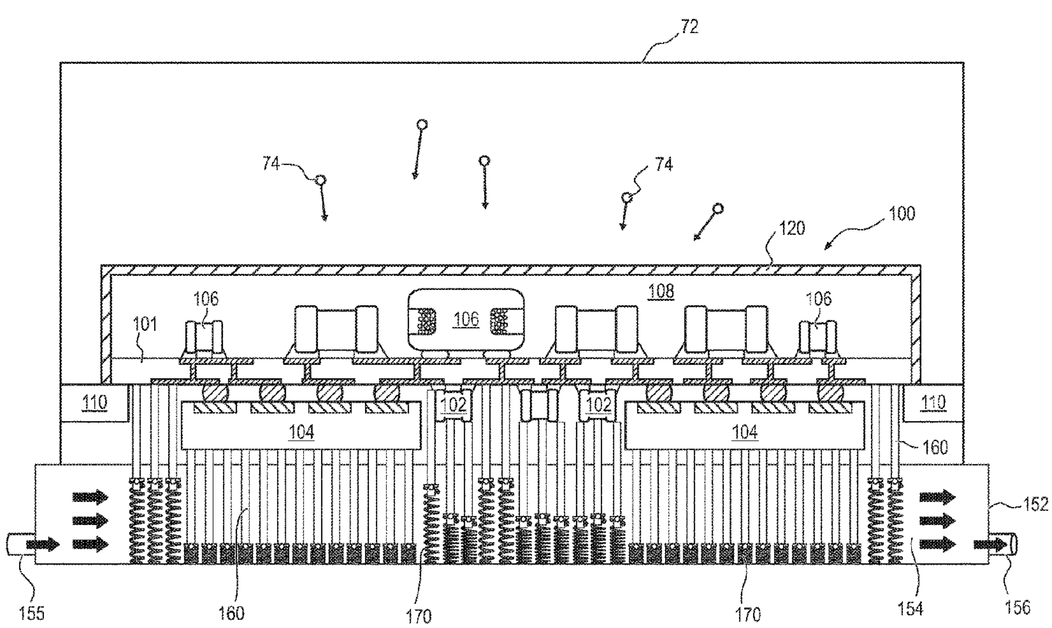
(57) **ABSTRACT**

A semiconductor manufacturing device has a cooling pad with a plurality of movable pins. The cooling pad includes a fluid pathway and a plurality of springs disposed in the fluid pathway. Each of the plurality of springs is disposed under a respective movable pin. A substrate includes an electrical component disposed over a surface of the substrate. The substrate is disposed over the cooling pad with the electrical component oriented toward the cooling pad. A force is applied to the substrate to compress the springs. At least one of the movable pins contacts the substrate. A cooling fluid is disposed through the fluid pathway.

24 Claims, 7 Drawing Sheets



Related U.S. Application Data

division of application No. 17/032,437, filed on Sep. 25, 2020, now Pat. No. 11,434,561.

(60) Provisional application No. 63/001,213, filed on Mar. 27, 2020.

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H01L 23/00	(2006.01)
H01L 23/498	(2006.01)
H01L 23/552	(2006.01)

(52) **U.S. Cl.**

CPC ***H01L 23/552*** (2013.01); ***H01L 24/16*** (2013.01); *H01L 2224/16227* (2013.01); *H01L 2924/19105* (2013.01); *H01L 2924/19106* (2013.01); *H01L 2924/3025* (2013.01)

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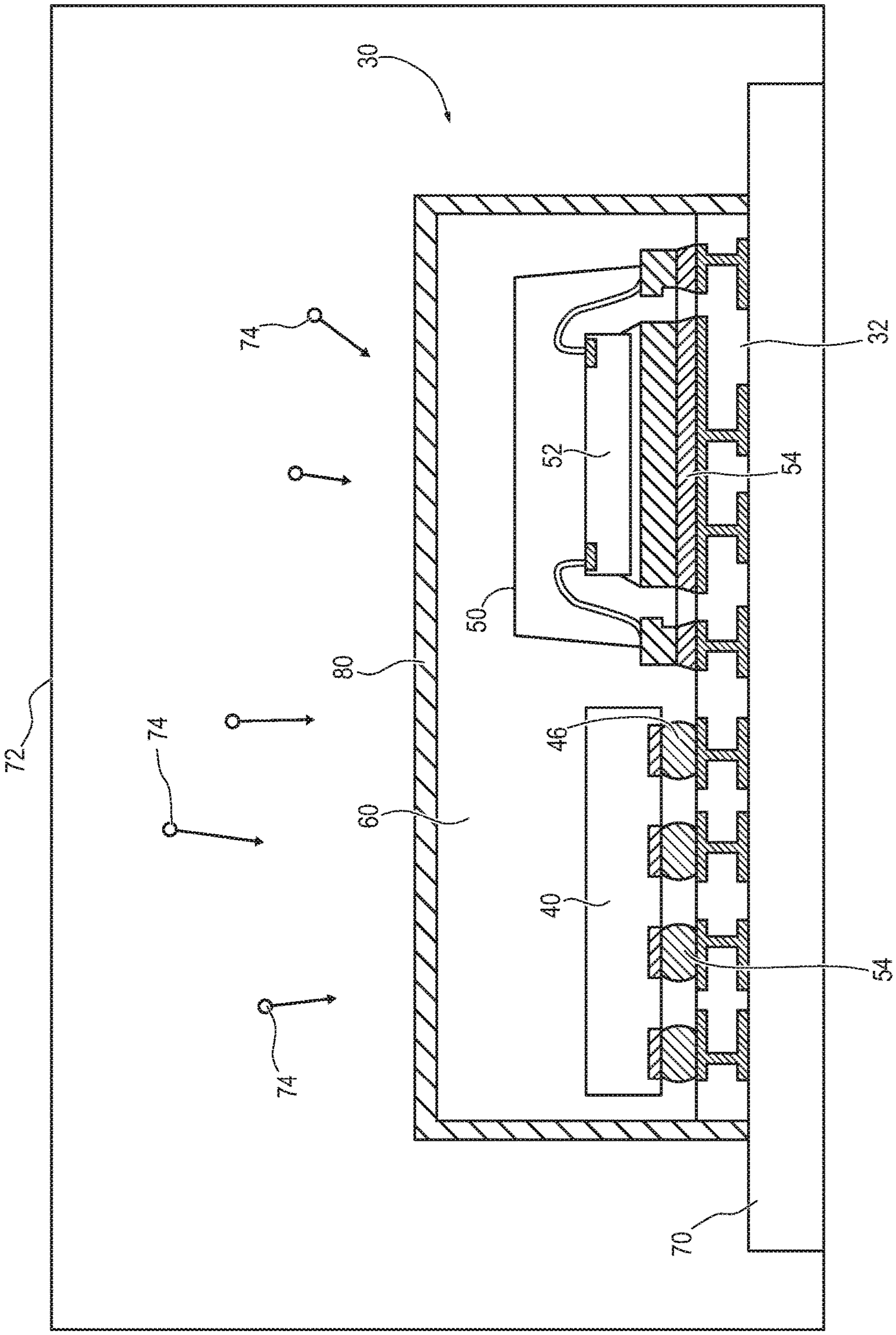


FIG. 1
(Prior Art)

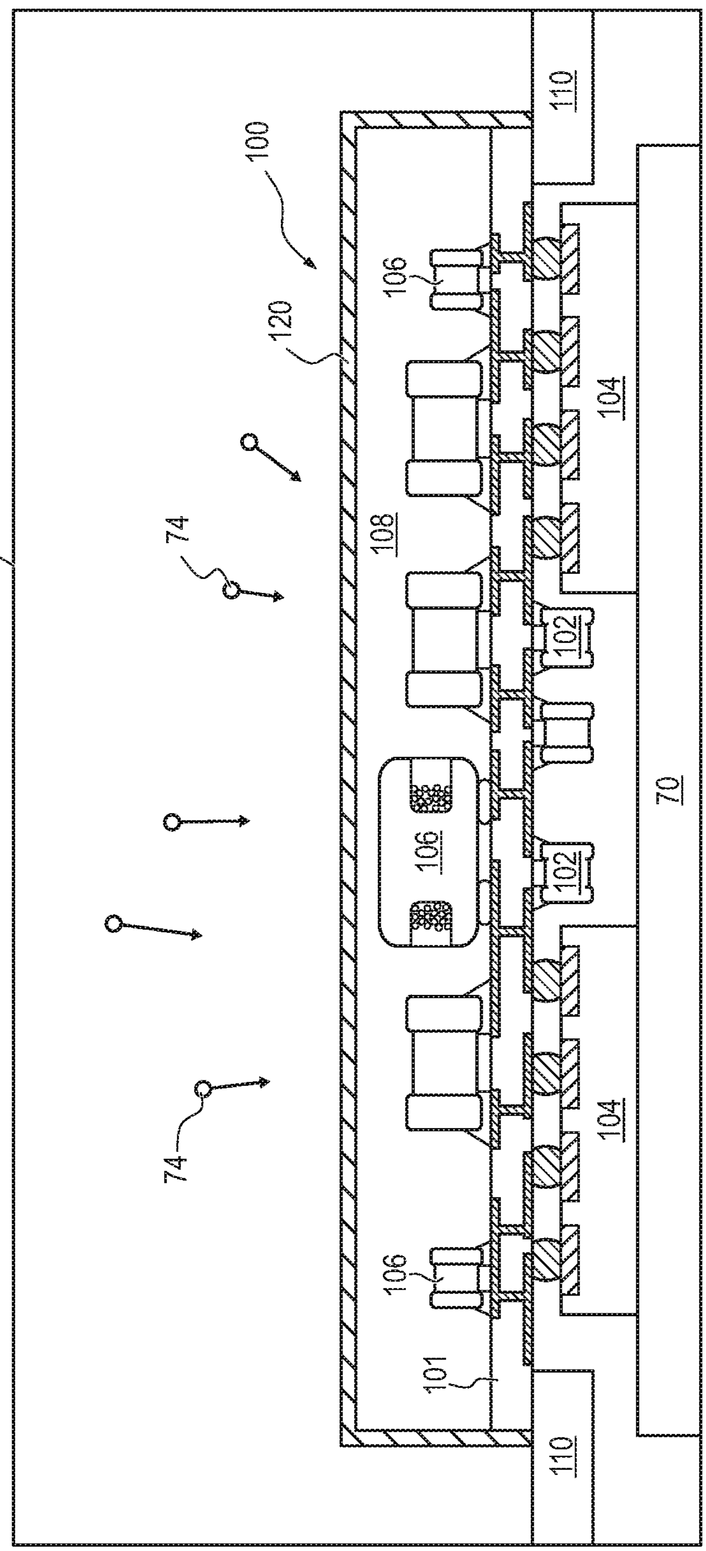


FIG. 2
(Prior Art)

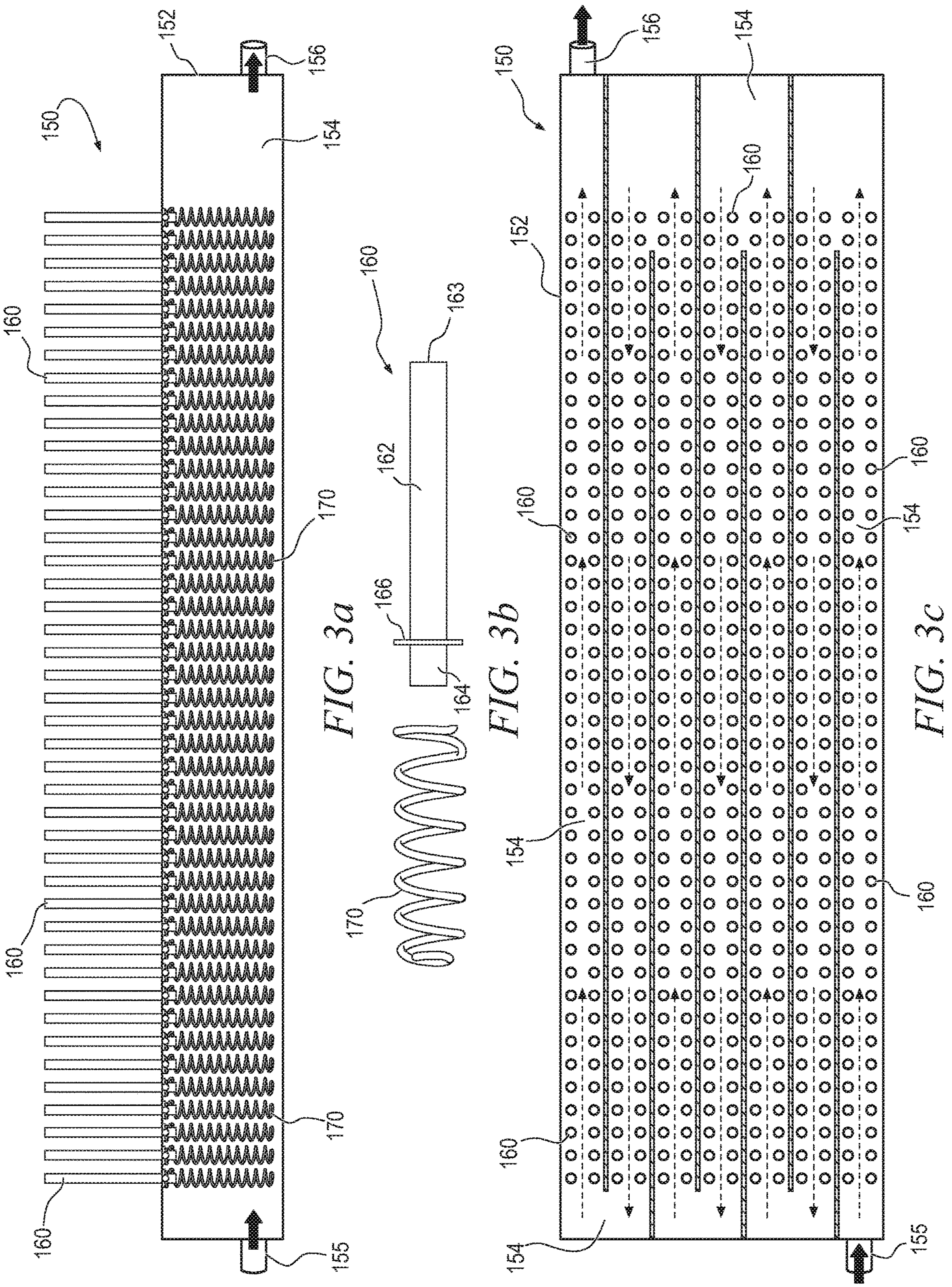
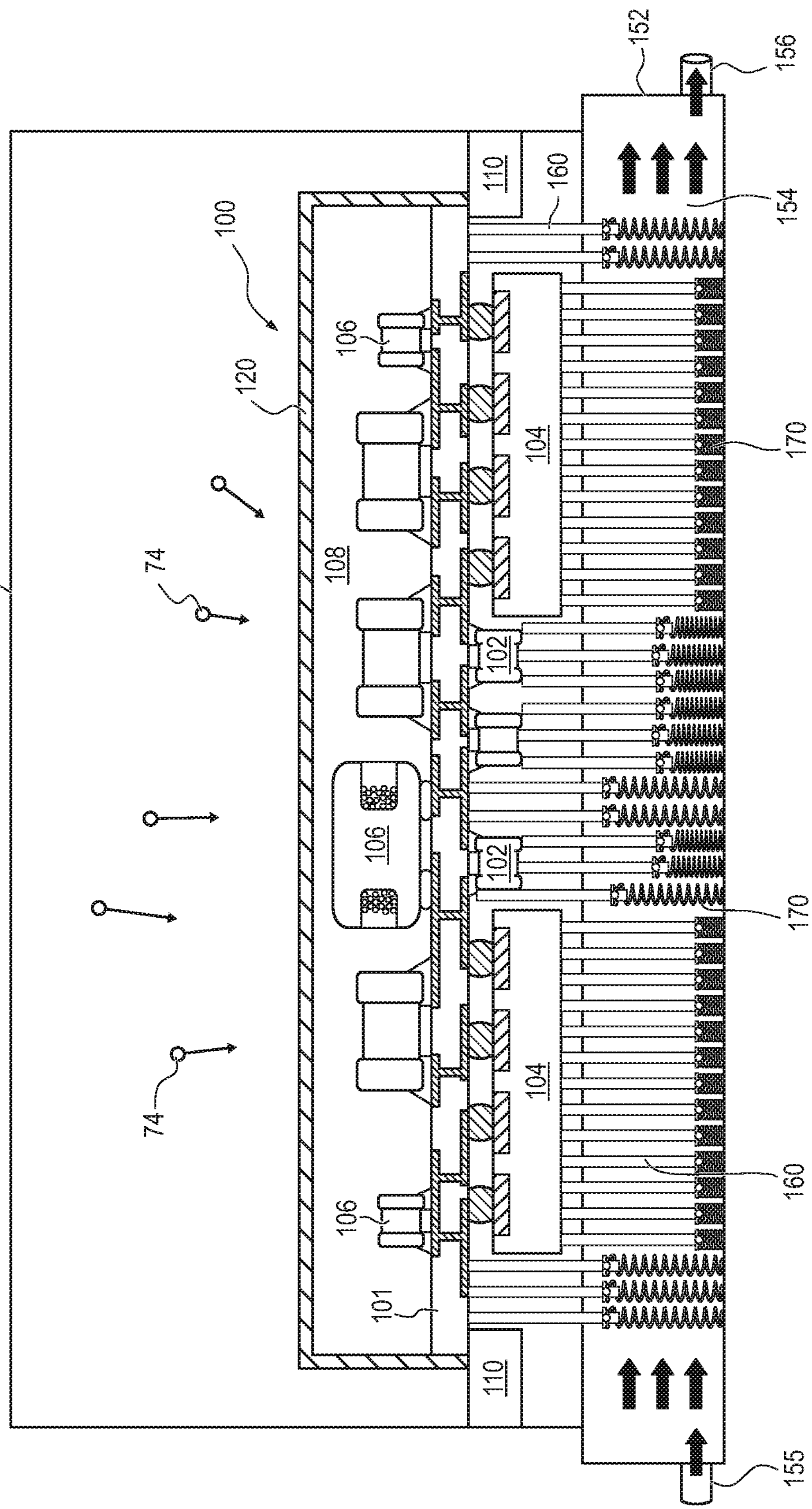


FIG. 3a

FIG. 3b

FIG. 3c


72
74
74
100
120
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110
110
160
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152
154
156
155
170
170
101

FIG. 4

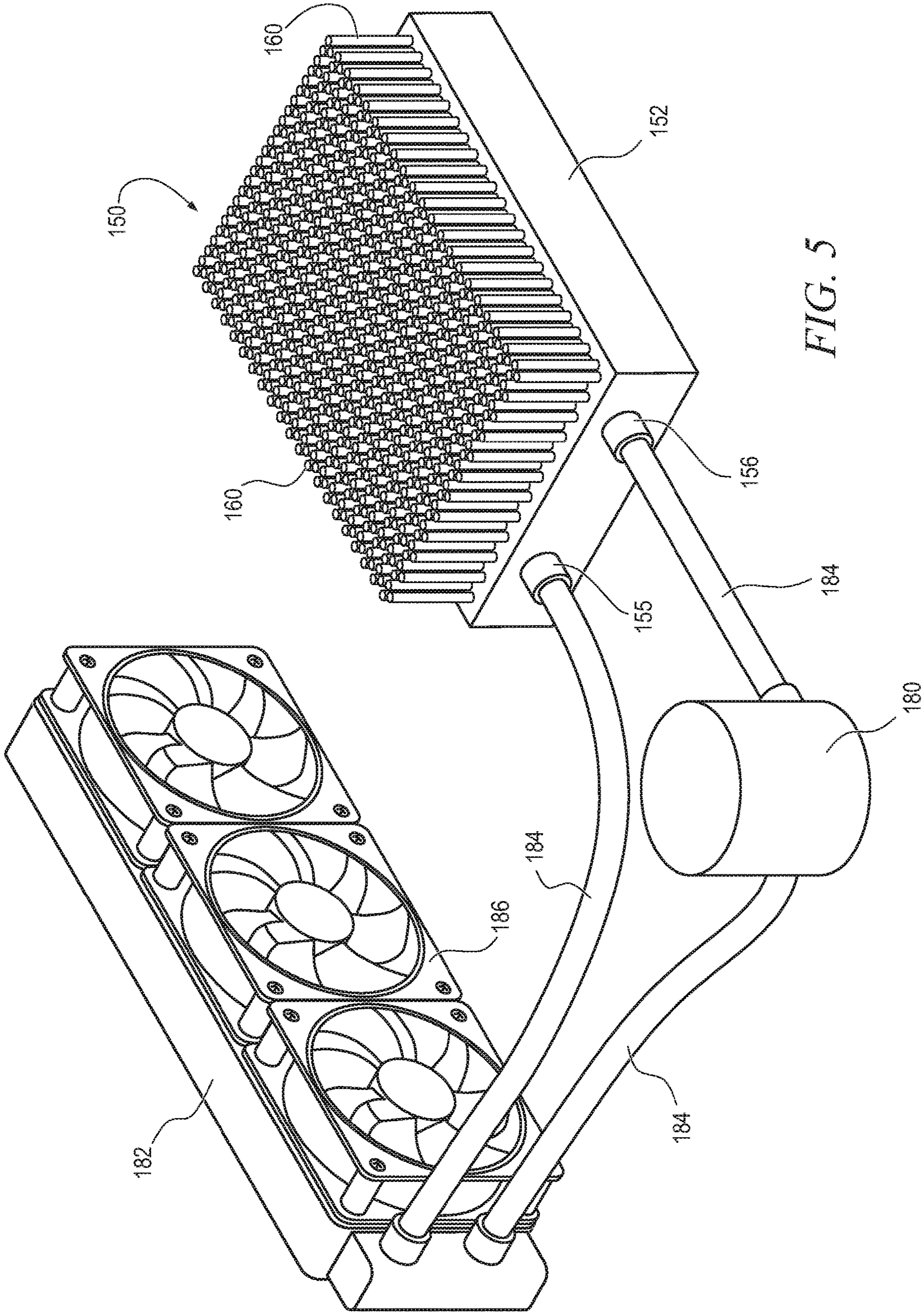

FIG. 5

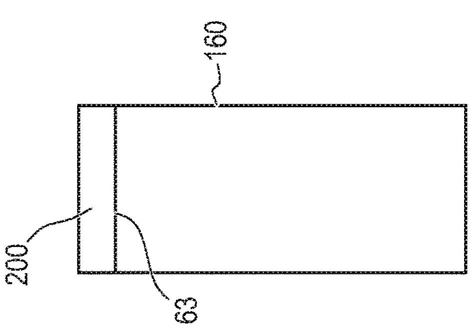

FIG. 6

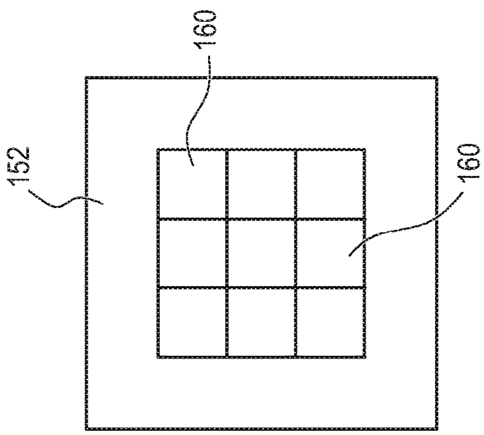

FIG. 7a

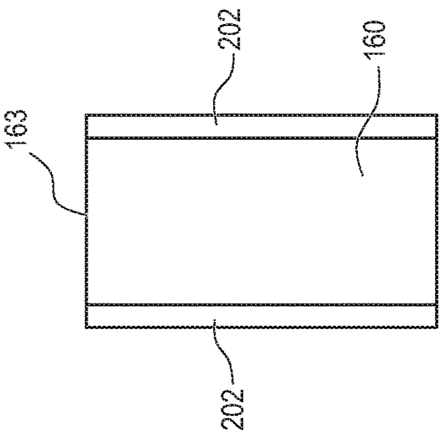

FIG. 7c

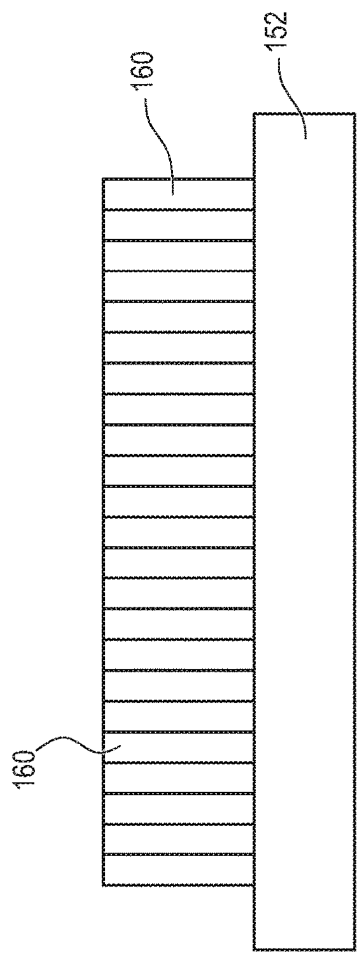

FIG. 7b

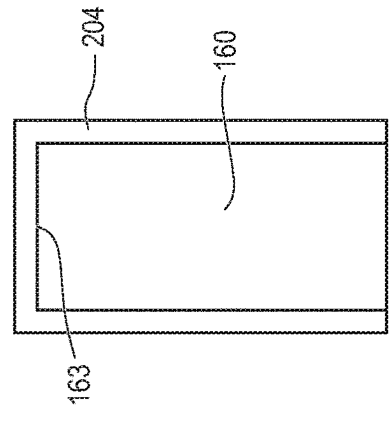

FIG. 7d

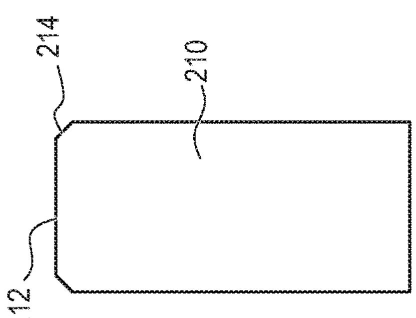

FIG. 8a

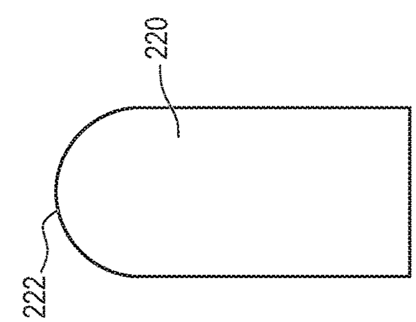

FIG. 8b

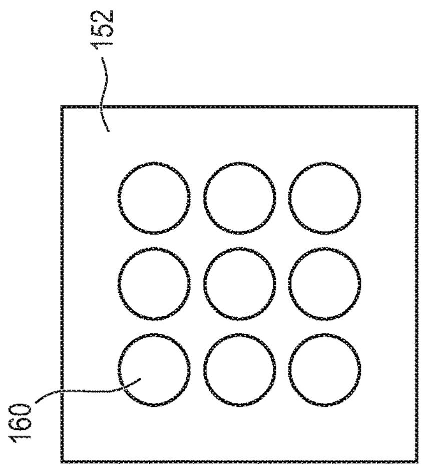

FIG. 9a

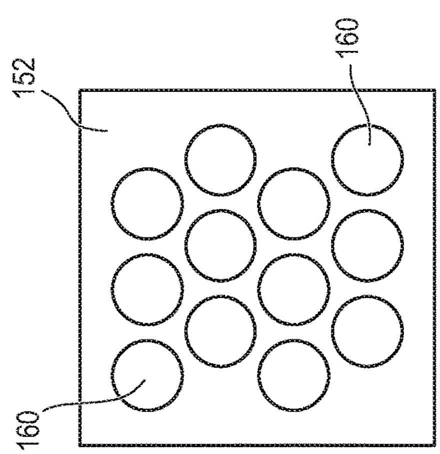

FIG. 9b

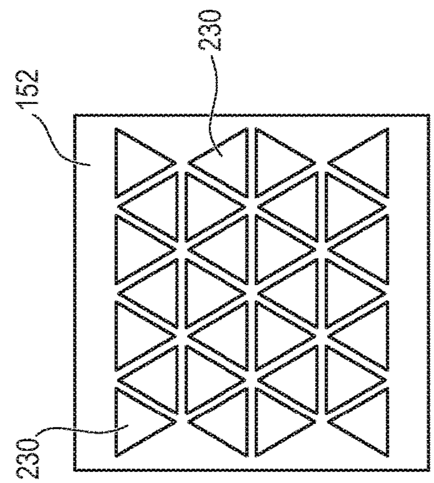

FIG. 9c

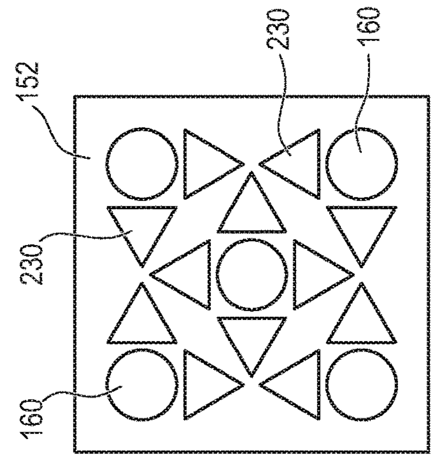

FIG. 9d

COOLING DEVICE AND PROCESS FOR COOLING DOUBLE-SIDED SIP DEVICES DURING SPUTTERING

CLAIM TO DOMESTIC PRIORITY

The present application is a continuation of U.S. patent application Ser. No. 17/814,796, now U.S. Pat. No. 11,932, 933, filed Jul. 25, 2022, which is a division of U.S. patent application Ser. No. 17/032,437, now U.S. Pat. No. 11,434, 561, filed Sep. 25, 2020, which claims the benefit of priority of U.S. Provisional Application No. 63/001,213, filed Mar. 27, 2020, which applications are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention relates in general to semiconductor manufacturing and, more particularly, to a device and method for cooling double-sided system-in-package (SiP) devices during sputtering.

BACKGROUND OF THE INVENTION

Semiconductor devices are commonly found in modern electronic products. Semiconductor devices perform a wide range of functions such as signal processing, high-speed calculations, transmitting and receiving electromagnetic signals, controlling electronic devices, transforming sunlight to electricity, and creating visual images for television displays. Semiconductor devices are found in the fields of communications, power conversion, networks, computers, entertainment, and consumer products. Semiconductor devices are also found in military applications, aviation, automotive, industrial controllers, and office equipment.

Semiconductor devices are often susceptible to electromagnetic interference (EMI), radio frequency interference (RFI), harmonic distortion, or other inter-device interference, such as capacitive, inductive, or conductive coupling, also known as cross-talk, which can interfere with their operation. The high-speed switching of digital circuits also generates interference.

Conductive layers can be formed over semiconductor packages to shield electronic parts within the package from EMI and other interference. Shielding layers absorb EMI before the signals can hit semiconductor die and discrete components within the package, which might otherwise cause malfunction of the device. Shielding layers are also formed over packages with components that are expected to generate EMI to protect nearby devices.

The shielding layers are commonly formed by sputtering, which generates a significant amount of heat. Unfortunately, increasing the temperature of a package during sputtering can cause several problems, such as remelting of solder, extrusion, warpage, or material damage. Therefore, semiconductor packages are typically disposed on a cooling pad during sputtering to keep the package under 150-200 degrees Celsius (° C.). Staying below 200° C. is generally satisfactory, but staying below 150° C. is preferred.

FIG. **1** illustrates a semiconductor package **30** being sputtered to add a shielding layer. Package **30** includes a package substrate **32**. A semiconductor die **40**, sub-package **50** with semiconductor die **52**, and other surface mount components are disposed on substrate **32** to provide the electrical functionality of package **30**. Solder **54** is used to physically and electrically couple semiconductor die **40** and sub-package **50** to substrate **32**. Melting of solder **54** during sputtering could cause discontinuities in the electrical connections to substrate **32**.

An encapsulant or molding compound **60** is deposited over substrate **32**, die **40**, and sub-package **50**. After encapsulation, package **30** is disposed on cooling pad **70** within sputtering machine **72**. Package **30** is bombarded with metal molecules **74**, e.g. copper, within sputtering machine **72** to build up a conductive shielding layer **80**. The temperature within sputtering machine **72** stays around 400° C. during sputtering, and thermal energy is constantly added to package **30** while sputtering is ongoing. Substrate **32** lies flat on cooling pad **70** for good thermal contact between package **30** and the cooling pad. Cooling pad **70** is made of a flexible material to tightly adhere to the bottom surface of substrate **32**. Cooling pad **70** withdraws thermal energy through substrate **32** to keep package **30** below a desired target temperature.

Newer packaging types, such as system-in-package devices, commonly use surface mount components on both sides of a package substrate as shown with package **100** in FIG. **2**. Substrate **101** of package **100** has passive components **102** and semiconductor die **104** mounted on the bottom surface and passive components **106** mounted on the top surface. Any combination of components can be disposed on any surface of substrate **101**. Encapsulant **108** is deposited over substrate **101** and components **106**. A frame **110** holds package **100** in sputtering machine **72** while the sputtering process deposits shielding layer **120** over encapsulant **108**.

As with package **30** in FIG. **1**, package **100** is disposed on cooling pad **70**. However, due to components **102** and **104** on the bottom surface, substrate **101** does not directly contact cooling pad **70**. Thermal energy can only be withdrawn from components **106** via semiconductor die **104**. Not having direct contact with substrate **101** reduces the rate of thermal energy being extracted from the top part of package **100** and means that the temperature of the package is unlikely to remain satisfactory. Package **100** has a high likelihood of manufacturing defects being introduced during sputtering because of the elevated temperature. Therefore, a need exists for an improved cooling mechanism usable with double-sided SiP devices.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates sputtering a semiconductor package;

FIG. **2** illustrates sputtering a double-side system-in-package device;

FIGS. **3***a*-**3***c* illustrate a cooling pad having movable pins;

FIG. **4** illustrates the double-sided system-in-package device in a sputtering machine with the cooling pad having movable pins;

FIG. **5** illustrates a cooling circuit including the cooling pad having movable pins;

FIG. **6** illustrates a coating formed on top of the movable pins;

FIGS. **7***a*-**7***d* illustrate movable pins disposed in physical contact with each other and a coating on the sides of the movable pins to reduce friction;

FIGS. **8***a* and **8***b* illustrate different shapes for the tips of the movable pins; and

FIGS. **9***a*-**9***d* illustrate different shapes for the footprints of the movable pins.

DETAILED DESCRIPTION OF THE DRAWINGS

The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. The term "semiconductor die" as used herein refers to both the singular and plural form of the words, and accordingly, can refer to both a single semiconductor device and multiple semiconductor devices.

Semiconductor devices are generally manufactured using two complex manufacturing processes: front-end manufacturing and back-end manufacturing. Front-end manufacturing involves the formation of a plurality of die on the surface of a semiconductor wafer. Each die on the wafer contains active and passive electrical components, which are electrically connected to form functional electrical circuits. Active electrical components, such as transistors and diodes, have the ability to control the flow of electrical current. Passive electrical components, such as capacitors, inductors, and resistors, create a relationship between voltage and current necessary to perform electrical circuit functions.

Back-end manufacturing refers to cutting or singulating the finished wafer into the individual semiconductor die and packaging the semiconductor die for structural support, electrical interconnect, and environmental isolation. To singulate the semiconductor die, the wafer is scored and broken along non-functional regions of the wafer called saw streets or scribes. The wafer is singulated using a laser cutting tool or saw blade. After singulation, the individual semiconductor die are mounted to a package substrate that includes pins or contact pads for interconnection with other system components. Contact pads formed over the semiconductor die are then connected to contact pads within the package. The electrical connections can be made with conductive layers, bumps, stud bumps, conductive paste, or wirebonds. An encapsulant or other molding compound is deposited over the package to provide physical support and electrical isolation. The finished package is then inserted into an electrical system and the functionality of the semiconductor device is made available to the other system components.

Electromagnetic interference (EMI) shielding layers are commonly formed over semiconductor packages as part of back-end manufacturing. As explained above, sputtering machines used to form the shielding layers generate heat that can cause manufacturing defects. FIG. **3***a* illustrates a cooling pad **150** with movable pins **160** that can be used in a sputtering machine to keep double-sided system-in-package (SiP) devices sufficiently cool. Cooling pad **150** has a base **152** with a fluid pathway **154** running through the base. During use, a cooling fluid flows through fluid pathway **154** to carry away thermal energy. The cooling fluid can be a chemical coolant, refrigerant, water, oil, gas, or any other suitable fluid.

Pins **160** extend through openings in the top of base **152** so that the top of the pins are outside of the base and the bottoms of the pins are within fluid pathway **154**. The opening in base **152** are just large enough to allow pins **160** to move in the openings without leaking fluid through the openings. In some embodiments, a gromet or other mechanism is used to seal the opening around pins **160**. Pins **160** are formed from a metal, such as copper, aluminum, or gold, or a polymer having high thermal conductivity, or another material having suitable thermal conductivity.

Pins **160** are spring-loaded with springs **170** so that the pins can be pressed into fluid pathway **154** by a package to be cooled. Springs **170** are formed from any suitable material, such as those mentioned above for pins **160**. Springs **170** can be made of a thermally conductive material to aid in thermal transfer to the cooling fluid. After the package is removed, pins **160** spring back out to the position shown in FIG. **3***a*. In some embodiments, an elastic material that stretches is used instead of coiled springs. Any suitable elastic mechanism can be used to return pins **160** to their fully extended position.

FIG. **3***b* shows a pin **160** and spring **170** separated from base **152**. Pin **160** includes a top portion **162** with a tip **163**. Top portion **162** is the portion of pin **160** that extends outside of base **152**. A bottom portion **164** of pin **160** is the portion that is disposed within fluid pathway **154**. Bottom portion **164** extends into spring **170** to keep the spring on the pin within base **152**. In one embodiment, top portion **162** and bottom portion **164** are of a uniform cross-section when viewed from tip **163**. In other embodiments, bottom portion **164** is thinner, thicker, or differently shaped than top portion **162** and may be simply a small bump on the bottom of the pin. In some embodiments, spring **170** extends into a groove on the bottom of pin **160** rather than around a bump. Fluid pathway **154** optionally has bumps or other structures on the bottom of the fluid pathway for holding the opposite ends of springs **170**.

Pin **160** includes a flange **166** formed around the pin between top portion **162** and bottom portion **164**. Flange **166** serves dual purposes, to both keep pin **160** within base **152** and to allow the pin to interact with spring **170**. When a package to be cooled presses down on tip **163** of pin **160**, the pin moves downward into base **152**. Flange **166** presses against the top of spring **170** as pin **160** moves downward, thereby compressing the spring. When the package is removed from cooling pad **150**, springs **170** decompress and press against flange **166** to move the pin back upward. Eventually, flange **166** hits the top of fluid pathway **154** and springs **170** stop decompressing. Flange **166** keeps spring **170** from undesirably pushing pin **160** completely out of base **152**.

Flange **166** is a disk having a common center with top portion **162** and bottom portion **164**. Flange **166** has a circular shape with a circumference extending completely around pin **160**. In other embodiments, flange **166** is simply two dowels or bumps extending in opposite directions from the pin. One dowel or bump on only one side of pin **160** is used in other embodiments. Having a flange **166** that does not extend in all directions from the center of pin **160** allows adjacent pins to be disposed closer together. The discrete flange **166** portions of adjacent pins **160** can be positioned offset from each other so that the pins can be formed within the distance of a single flange width of each other, rather than having to be spaced by two flange widths. No flange is necessary in embodiments where spring **170** extends into a groove on the bottom of pin **160**, or if the springs otherwise apply force to the bottom of the pins. A tab of base **152** can extend into pin **160** to keep the pins within the base, rather than relying on flange **166** hitting the top of fluid pathway **154**.

FIG. **3***c* illustrates a top-down plan view of cooling pad **150** with base **152** cross-sectioned to show fluid pathway **154** with a serpentine shape. The footprint of cooling pad **150** can be expanded to accommodate any sized semiconductor package, or a multi-device panel that will be singulated after sputtering. Cooling fluid flows back and forth across the entire width of base **152** multiple times to get from fluid inlet **155** to fluid outlet **156**. The serpentine shape of fluid pathway **154** forces the cooling fluid to flow across each pin **160** and spring **170** with substantially equal volume per unit time. In other embodiments, fluid pathway **154** is one large chamber, multiple parallel paths, or any other suitable shape.

FIG. **4** shows cooling pad **150** disposed in sputtering machine **72** with double-sided SiP device **100** disposed on the cooling pad. SiP device **100** is pressed down onto pins **160**. Tips **163** of pins **160** press against SiP device **100** as the SiP device is pressed downward. Each pin **160** is pressed down a certain distance depending on what portion of SiP device **100** is directly over the particular pin. Pins **160** directly under die **104** are pressed down the most, such that springs **170** are fully or nearly fully compressed. Pins **160** that are directly under discrete components **102** are not pressed down as far as pins under die **104** but still have tips **163** that are pressed against the discrete components by springs **170**.

Pins **160** that are not under any components on the bottom of substrate **101** have tips **163** pressing against the bottom surface of the substrate. The pins **160** that contact substrate **101** still compress respective springs **170** somewhat so that the pins apply some force against the substrate. In other embodiments, substrate **101** is positioned to rest in contact with pins **160** without compressing respective springs **170**.

Pins **160** draw thermal energy from SiP device **100** through physical contact. Because the heights of pins **160** adjust based on the shape of the bottom of a package being sputtered, the pins provide physical contact of the cooling surface for a much larger surface area than prior art cooling pads. Pins **160** not only draw thermal energy from die **104**, but also directly from substrate **101** and discrete components **102**.

Pins **160** transfer thermal energy down to within fluid pathway **154** where the thermal energy is further transferred to the cooling fluid and carried away by the flow. Thermal energy is also transferred from pins **160** to springs **170**, which helps present a larger surface area to the flow of cooling fluid due to the coiled shape. All pins contact the bottom of package **100**, which provides sufficient thermal capacity to keep the package under 150-200° C. In one embodiment, top portion **162** of each pin is made at least as long, from flange **166** to tip **163**, as the tallest expected device to be disposed on the bottom side of a package being sputtered so that in all expected cases pins **160** are able to physically contact substrate **101**.

FIG. **5** illustrates a full cooling loop with cooling pad **150**, pump **180**, and radiator **182** coupled in a fluid circuit via tubes **184**. Pump **180** creates a pressure differential between an input and output of the pump to propel fluid through the system. In the illustrated configuration, pump **180** pulls cooling fluid from base **152** via outlet **156** and a first tube **184**, and then pushes the fluid into radiator **182** via a second tube **184**. The fluid flows through radiator **182** and then back to inlet **155** of cooling pad **150** via a third tube **184**. Radiator **182** has fans **186** attached to push air through the radiator and cool off the fluid. Radiator **182** includes a fluid pathway and several fins that are warmed by the cooling fluid. Fans **186** blow ambient air through the radiator across the fins to transfer thermal energy from the fins to the ambient air. Cooling fluid is returned to base **152** via tube **184** to absorb more thermal energy from package **100** via pins **160**.

Any type of thermal heat exchanger can be used instead of a radiator as illustrated. In some embodiments, electrically powered Peltier coolers are used. One embodiment utilizes a refrigeration cycle to cool base **152**. The refrigeration cycle uses a refrigerant as the cooling fluid, and has a compressor, condenser, and expansion valve as part of the fluid circuit. Base **152** acts like an evaporator for the system and cools off pins **160** as part of the evaporation process. Base **152** may need a modified structure to properly act as an evaporator in the refrigeration cycle. Any type of heat exchangers can be used for radiator **182** and base **152**. Base **152** exchanges heat from package **100** to the cooling fluid. Radiator **182** exchanges heat from the cooling fluid to ambient air or another medium.

FIG. **6** illustrates a pin **160** with a surface coating **200** on tip **163** of the pin. Pins **160** of cooling pad **150** optionally have surface coating **200** made of a damping material, such as silicone or polytetrafluoroethylene (PTFE). Surface coating **200** reduces friction and softens tip **163**, which reduces the likelihood of damage to the device being cooled and helps accommodate slight lateral movements of the device relative to the pins.

In some embodiments, the gap between adjacent pins **160** becomes nearly zero, as shown in FIGS. **7***a* and **7***b*. FIG. **7***a* is a top-down plan view while FIG. **7***b* is a side view. For such embodiments, pins **160** are close enough that adjacent pins are likely to physically contact each other during use. FIG. **7***c* shows a coating **202** formed on the side surfaces all the way around pin **160**. Coating **202** can be formed from PTFE or another low-friction material that allows pins **160** to slide against each other. FIG. **7***d* shows a pin **160** with coating **204** formed both on the side surfaces and also on tip **163**. Coating **204** provides the benefits of both coating **200** and coating **202**. Alternatively, coating **200** and coating **202** can be formed separately from the same or two different materials on pins **160**.

Pins can be made with a wide variety of shapes. FIGS. **8***a* and **8***b* illustrate different tip shapes for pins. FIG. **8***a* shows pin **210** with a beveled tip **212**. A beveled surface **214** provides a transition from the side surface to the top surface. The edges of tip **212** can be rounded instead. FIG. **8***b* illustrates a pin **220** with the entire tip **222** rounded. Rounded or beveled tips reduce the likelihood of damage from sharp tip corners and help the pins conform to angled surfaces.

Pins can also be formed with a wide variety of profile shapes in plan view, as well as a variety of layouts within base **152**. FIG. **9***a* shows round pins **160** oriented in a regular grid on base **152**. FIG. **9***b* shows round pins **160** with rows of pins offset from each other. Rows being offset allows more round pins **160** to be placed in the same footprint area of base **152**. FIG. **9***c* shows pins **230** with triangular profile shapes. The triangular pins can have all three side surfaces of each pin facing another adjacent pin so that nearly the entire surface of base **152** is covered in pins.

Pins can have any desired profile shape in footprint view, e.g., rectangular, triangular, circular, etc. Multiple different shapes of pins can be used together on a single cooling pad, e.g., circular pins **160** and triangular pins **230** are used together in FIG. **9***d*. Each circular pin **160** is surrounded by a triangular pin **230** on each cardinal direction from the circular pin. The size, footprint shape, tip shape, and layout of pins is not limited.

While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims. While the description is drafted in terms of cooling during sputtering, the described cooling pad with movable pins can be used to cool any device in any situation.

What is claimed:

1. A method of making a semiconductor device, comprising:

providing a cooling pad including a fluid pathway formed in the cooling pad;

disposing a cooling fluid in the cooling pathway;

disposing a plurality of pins extending from within the fluid pathway to outside the cooling pad;

forming a coating on a top surface or a side surface of the plurality of pins; and

disposing a plurality of springs within the cooling pathway with a spring of the plurality of springs disposed under each of the plurality of pins.

2. The method of claim **1**, further including coupling a heat exchanger to the cooling pad.

3. The method of claim **1**, further including disposing a substrate over the cooling pad with a component of the substrate contacting one of the plurality of pins.

4. The method of claim **1**, further including disposing the substrate with a second of the plurality of pins contacting the substrate.

5. The method of claim **1**, further including forming the coating on the top surface of the plurality of pins.

6. The method of claim **1**, further including forming the coating on the side surface of the plurality of pins.

7. A method of making a semiconductor device, comprising:

providing a cooling pad including a fluid pathway formed in the cooling pad;

disposing a pin extending from within the fluid pathway to outside the cooling pad;

disposing a spring in the cooling pad under the pin; and

forming a coating on a surface of the pin.

8. The method of claim **7**, further including a heat exchanger coupled to the cooling pad.

9. The method of claim **7**, further including disposing a substrate over the cooling pad with a component of the substrate contacting the pin.

10. The method of claim **9**, further including disposing the substrate with a second pin extending from within the fluid pathway to physically contact the substrate.

11. The method of claim **7**, further including forming the coating on a top surface of the pin.

12. The method of claim **7**, further including forming the coating on a side surface of the pin.

13. The method of claim **7**, further including disposing a plurality of pins extending from within the fluid pathway to outside the cooling pad.

14. A method of making a semiconductor device, comprising:

providing a cooling pad including a fluid pathway formed in the cooling pad;

disposing a pin extending from within the fluid pathway to outside the cooling pad; and

forming a coating on a surface of the pin.

15. The method of claim **14**, further including a heat exchanger coupled to the cooling pad by a tube.

16. The method of claim **14**, further including disposing a substrate over the cooling pad with a component of the substrate contacting the pin.

17. The method of claim **16**, further including disposing the substrate with a second pin extending from within the fluid pathway to physically contact the substrate.

18. The method of claim **14**, further including disposing a plurality of pins extending from within the fluid pathway to outside the cooling pad.

19. A method of making a semiconductor device, comprising:

providing a cooling pad;

forming a fluid pathway in the cooling pad;

disposing a pin in the fluid pathway; and

forming a coating on a surface of the pin.

20. The method of claim **19**, further including coupling a heat exchanger to the cooling pad.

21. The method of claim **19**, further including disposing a substrate over the cooling pad with a component mounted on the substrate contacting the pin.

22. The method of claim **21**, further including disposing the substrate with a second pin extending from the fluid pathway to the substrate.

23. The method of claim **19**, further including disposing a plurality of pins in the fluid pathway.

24. The method of claim **1**, further including disposing a cooling fluid through the fluid pathway, wherein the cooling fluid flows in direct physical contact with the plurality of pins and plurality of springs.

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